# RETAIL SALES DATA ANALYSIS

## INTERNSHIP PROJECT REPORT

Submitted by: _____

Course / Semester: BCA _____

Institution: _____

Internship Organization: _____

Submission Date: _____

Project scope: Data cleaning, transformation, exploratory data analysis, customer segmentation, customer insights and visualization.

# Executive Summary

This internship project applies a complete Python-based workflow to a retail sales transaction dataset. The work covers six stages: data cleaning, data transformation, exploratory data analysis (EDA), customer segmentation, customer insights and data visualization. The purpose is to convert raw transaction records into reliable information that can support decisions about products, customers, inventory and sales timing.

The raw dataset contained 253 rows and 12 columns. The cleaning process removed 3 exact duplicate rows and 3 records with missing Customer_ID values, producing a final cleaned dataset of 247 rows and 12 columns. The analysis identified Wireless Mouse as the highest-quantity product (69 units) and Bluetooth Speaker as the highest-revenue product (■99,942). March 2025 was the strongest reported month (approximately ■70,345), while Friday was the strongest weekday. Frequent Buyers represented 51.8% of the customer segments.

# 1. Project Objectives

- Inspect and clean raw retail transaction data.
- Create derived fields and date-based variables.
- Explore sales, products, categories, customers and trends.
- Segment customers using frequency and monetary value.
- Translate findings into practical business recommendations.
- Present results through clear charts and a structured report.

# 2. Tools and Technologies

Python 3; Pandas; NumPy; Matplotlib; Seaborn; Scikit-learn (K-Means clustering); CSV datasets; Visual Studio Code.

# 3. Dataset Overview

The dataset contains Transaction_ID, Customer_ID, Customer_Name, Gender, Age, Product_ID, Product_Name, Category, Quantity, Price_Per_Unit, Total_Amount and Purchase_Date. Transformation generated additional fields including Total_Sales, Day_of_Week, Month, Year and Week_Number. The transformed dataset was reported to contain 17 columns.

# 4. Task 1 – Data Cleaning

The original dataset contained 253 rows and 12 columns. Inspection identified missing values in several fields and 3 exact duplicate rows. The cleaning workflow removed the 3 duplicates and 3 rows with missing Customer_ID values, handled other missing values with suitable imputation, standardized fields, checked data types and saved the result as retail_sales_cleaned.csv.

**Result:** 247 rows × 12 columns; the final missing-value check showed zero missing values in every column.

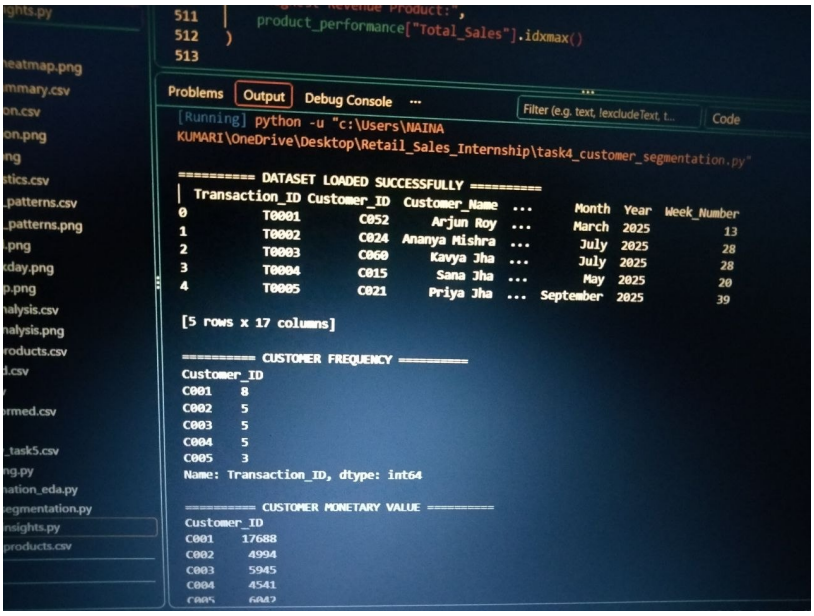



*Figure 1. Task 1 – Data-cleaning output shown in the Python console.*

# 5. Task 2 – Data Transformation

The cleaned data was enhanced for sales and time-based analysis. Sales values were calculated from Quantity × Price_Per_Unit; Purchase_Date was converted to datetime; Day_of_Week, Month, Year and Week_Number were extracted or calculated. The transformed dataset was saved as retail_sales_transformed.csv and contains 17 columns.

# 6. Task 3 – Exploratory Data Analysis

EDA summarized numerical variables and examined customer spending, product quantity, product revenue, monthly/weekly sales and category performance across time. Descriptive statistics were generated, and charts were used to compare distributions and trends. The analysis indicates that sales are not evenly distributed across products or time periods.

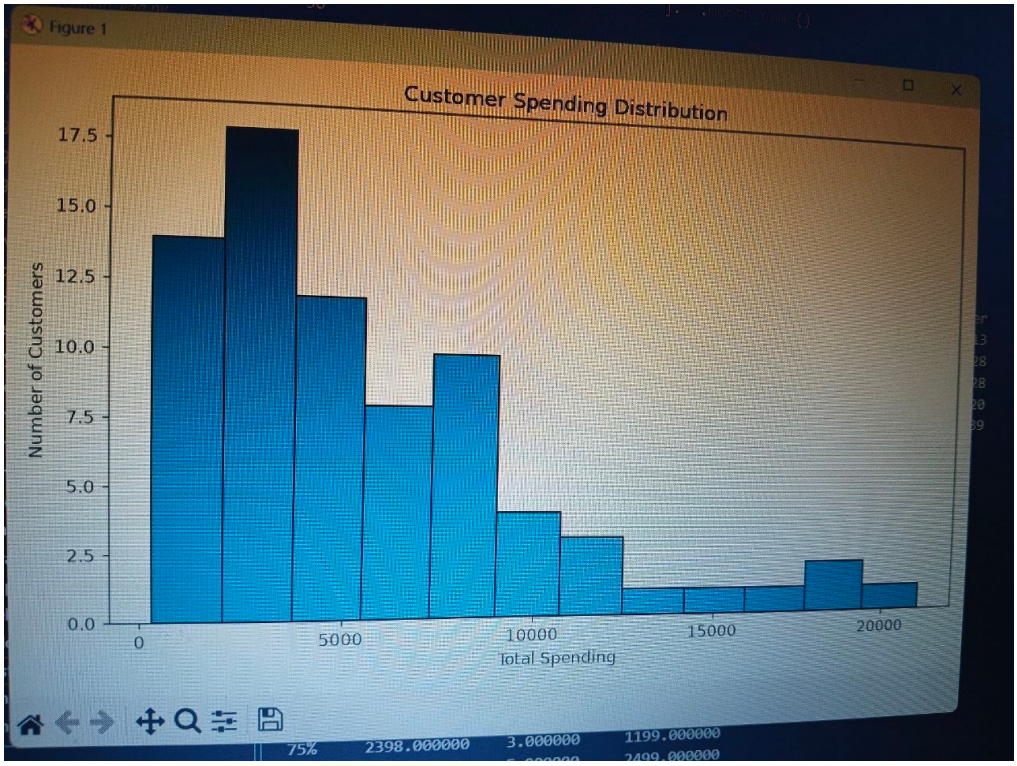



*Figure 2. Customer spending distribution – histogram.*

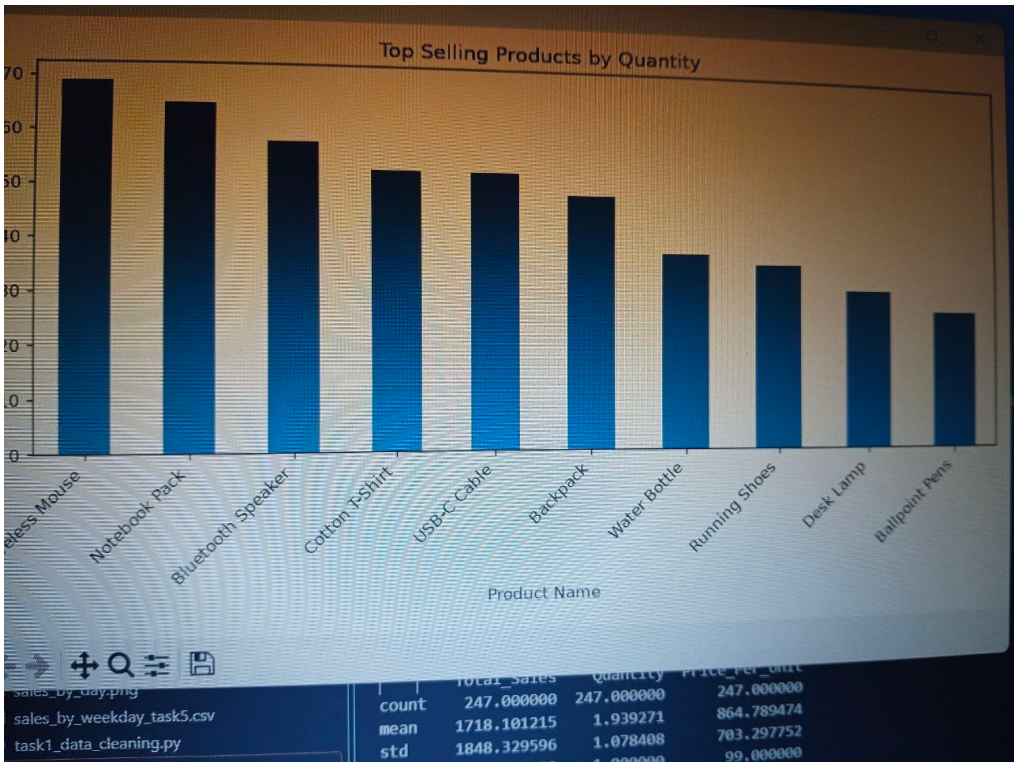



*Figure 3. Top-selling products ranked by quantity.*

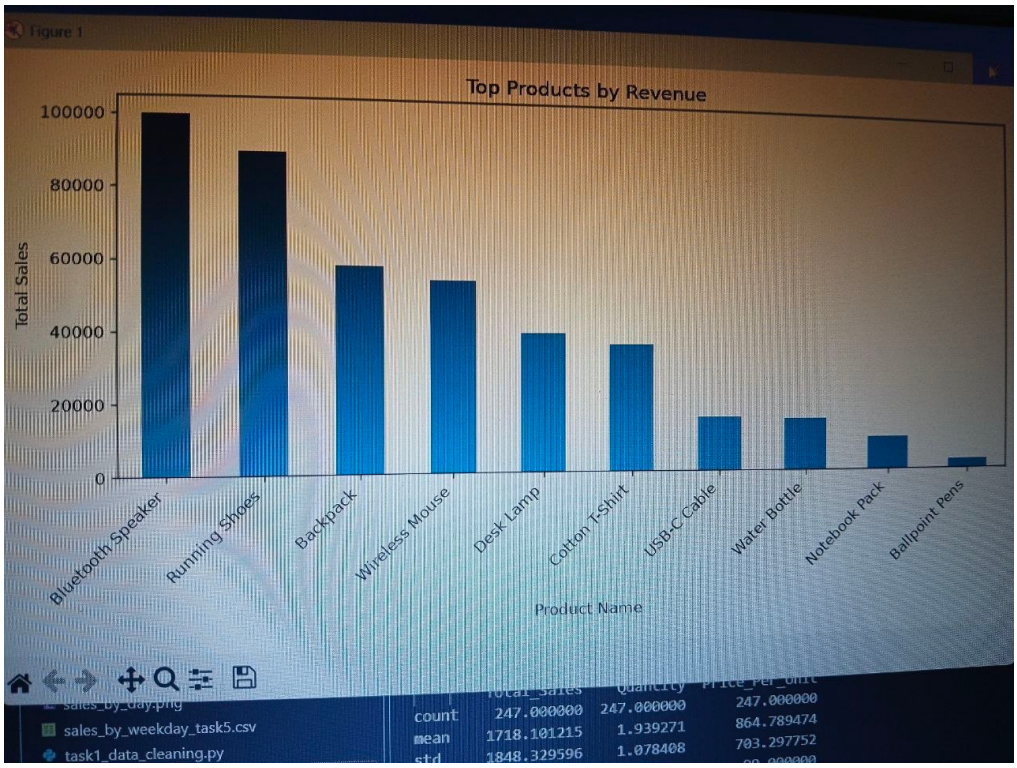



*Figure 4. Top products ranked by revenue.*

# 7. Task 4 – Customer Segmentation

Customers were grouped using Frequency (number of transactions) and Monetary Value (total customer spending). K-Means clustering was applied to form three groups: High Value Customers, Frequent Buyers and Infrequent / Low Value Customers. The segment chart reports Frequent Buyers at 51.8%, Infrequent / Low Value Customers at 25.9% and High Value Customers at 22.3%. Customer C001 is shown with 8 purchases and reported spending of ■17,688.

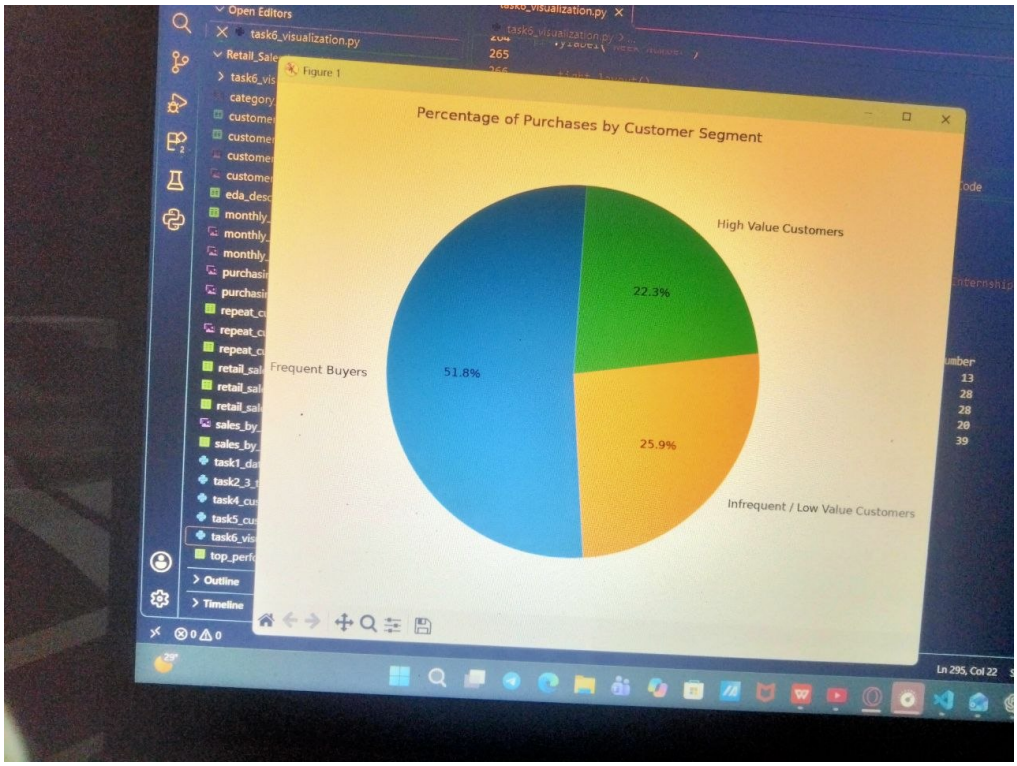



*Figure 5. Percentage distribution of purchases by customer segment.*

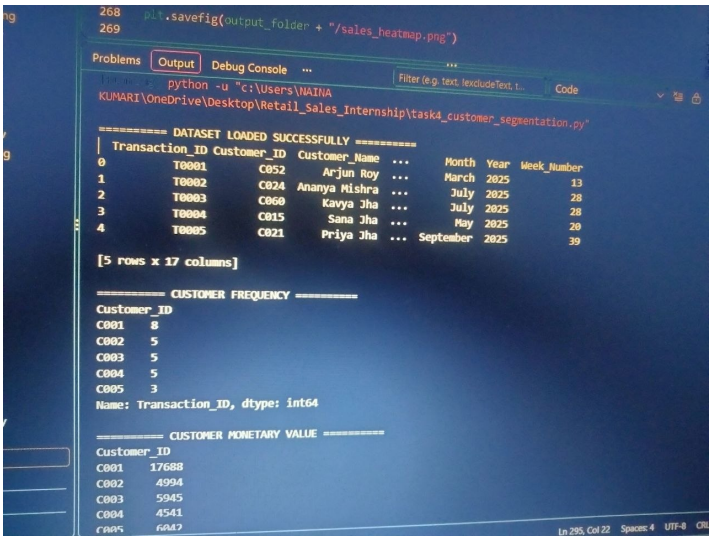



*Figure 6. Customer frequency and monetary-value output.*

# 8. Task 5 – Customer Insights and Recommendations

## Product performance

Wireless Mouse led by quantity sold, with 69 units. Bluetooth Speaker generated the highest reported product revenue, ■99,942. High-volume products require stock availability, while high-revenue products deserve attention to pricing, margins and promotion.

## Purchasing patterns

March 2025 was the strongest reported month at approximately ■70,345. Friday was the strongest weekday in the reported comparison. These patterns can guide campaign timing, staffing and inventory planning.

## Customer behaviour

Frequent Buyers formed the largest customer segment. Loyalty rewards and personalized offers may help retain frequent buyers, while reactivation campaigns may encourage infrequent customers to purchase again.

### Recommendations

- Maintain sufficient inventory for high-quantity products such as Wireless Mouse.
- Monitor revenue, cost and margin for high-revenue products such as Bluetooth Speaker.
- Plan promotions around stronger periods, including March-like seasonal peaks and Fridays.
- Provide loyalty benefits to frequent buyers and targeted offers to infrequent customers.
- Validate product bundles with transaction-level market-basket analysis before claiming specific cross-sell pairs.

## 9. Task 6 – Data Visualization

The visualization task uses a histogram, product bar charts, sales trend line plots, a customer-segment pie chart and a category-by-time heatmap to communicate the findings.

### Histogram – Customer spending

The histogram is right-skewed: most customers fall within lower-to-middle spending ranges, while a smaller group spends substantially more.

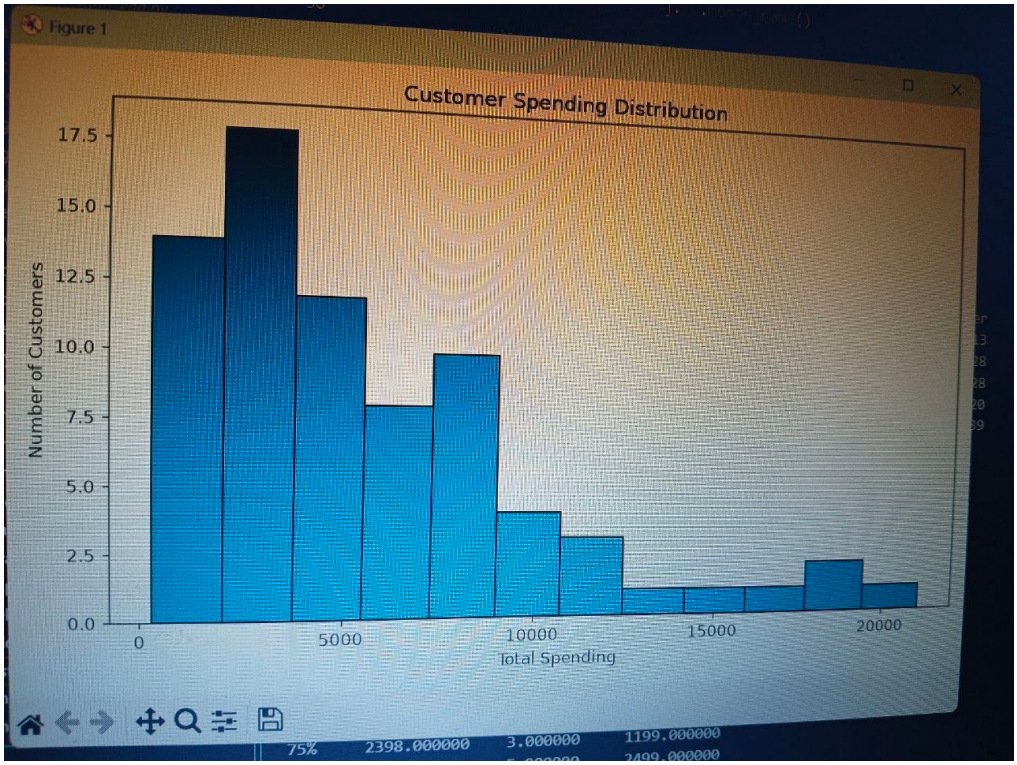



*Figure 7. Customer spending distribution.*

### Bar charts – Product performance

The quantity chart highlights frequently purchased products, while the revenue chart identifies products contributing the greatest sales value.

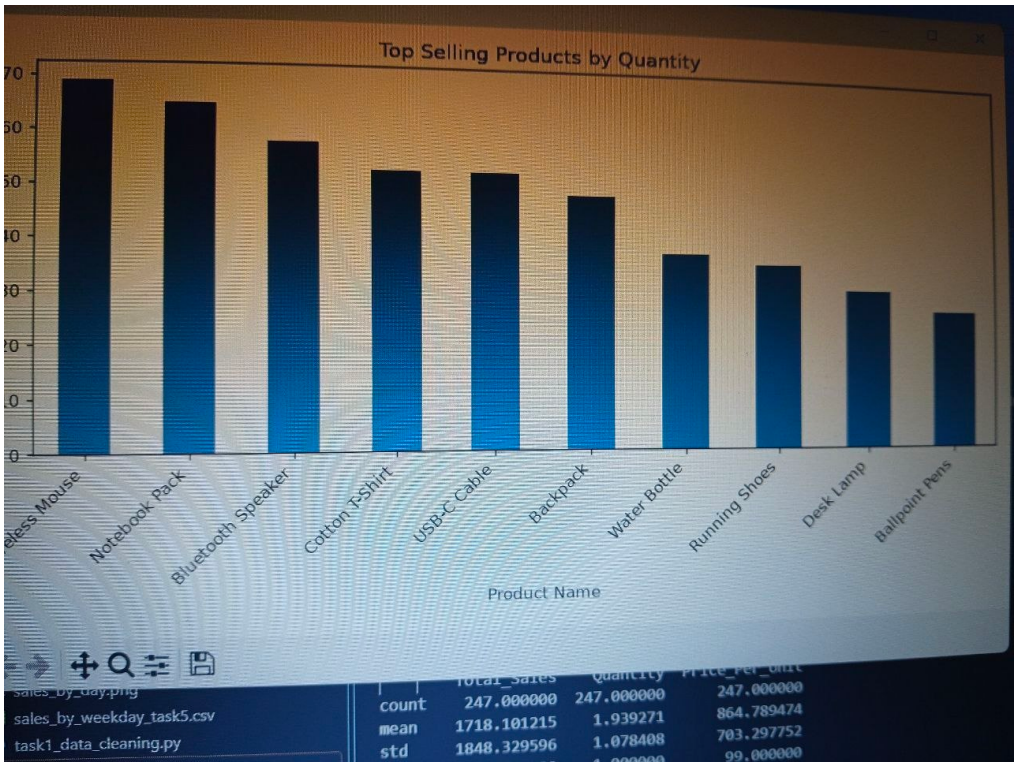



*Figure 8. Top-selling products by quantity.*

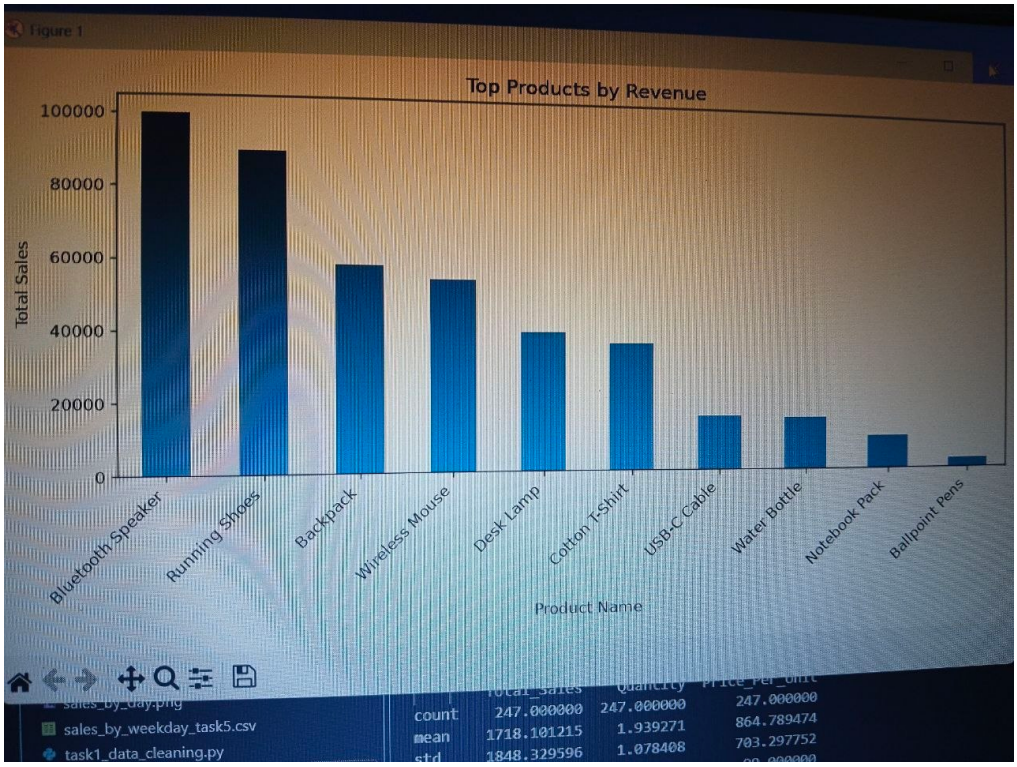



*Figure 9. Top products by revenue.*

## Line plots – Sales trends

The monthly line plot shows a visible peak in March 2025 and variation across the year. The weekly line plot shows substantial week-to-week fluctuations.

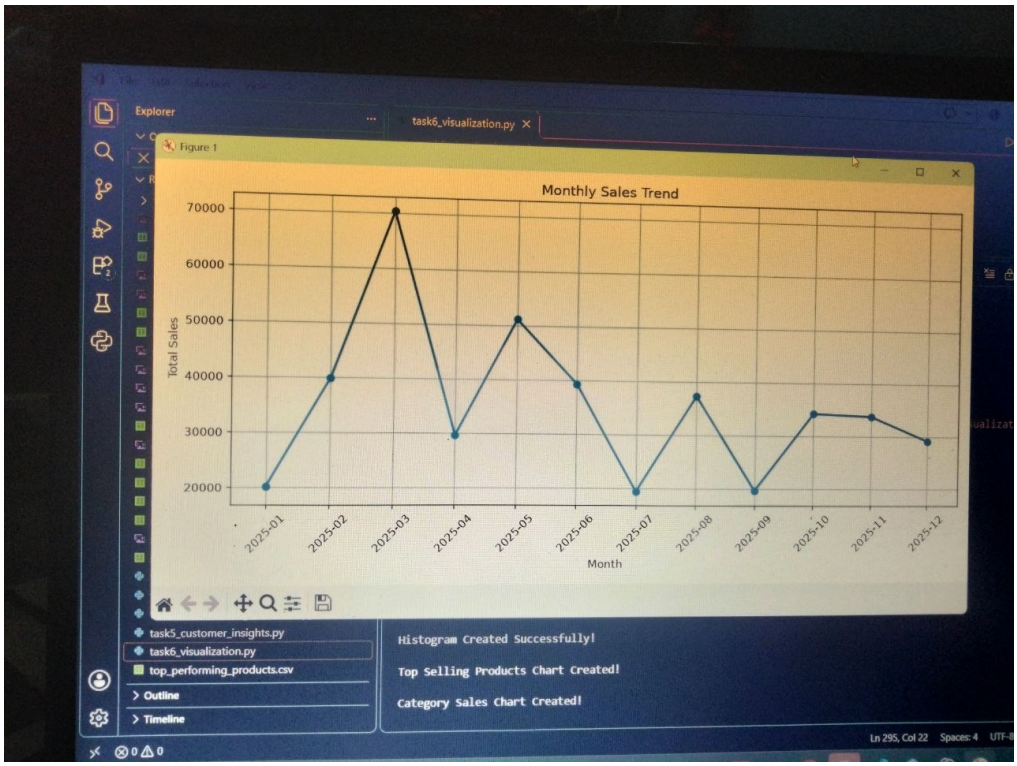



*Figure 10. Monthly sales trend.*

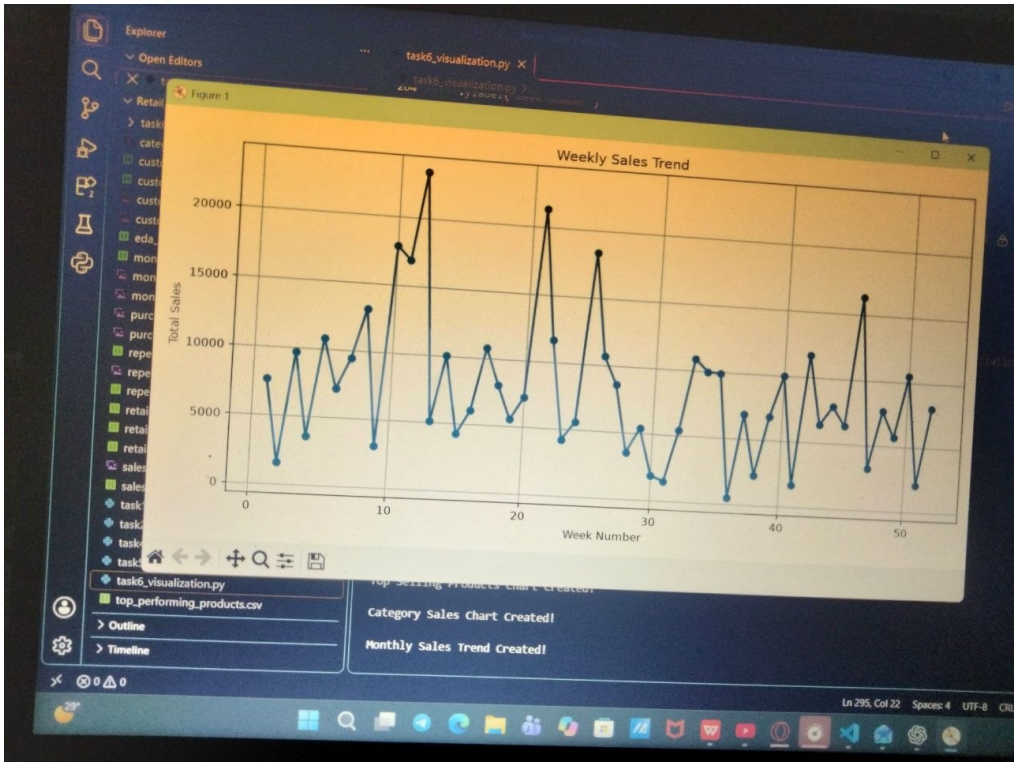



*Figure 11. Weekly sales trend.*

## Pie chart – Customer segments

The pie chart shows Frequent Buyers (51.8%), Infrequent / Low Value Customers (25.9%) and High Value Customers (22.3%).

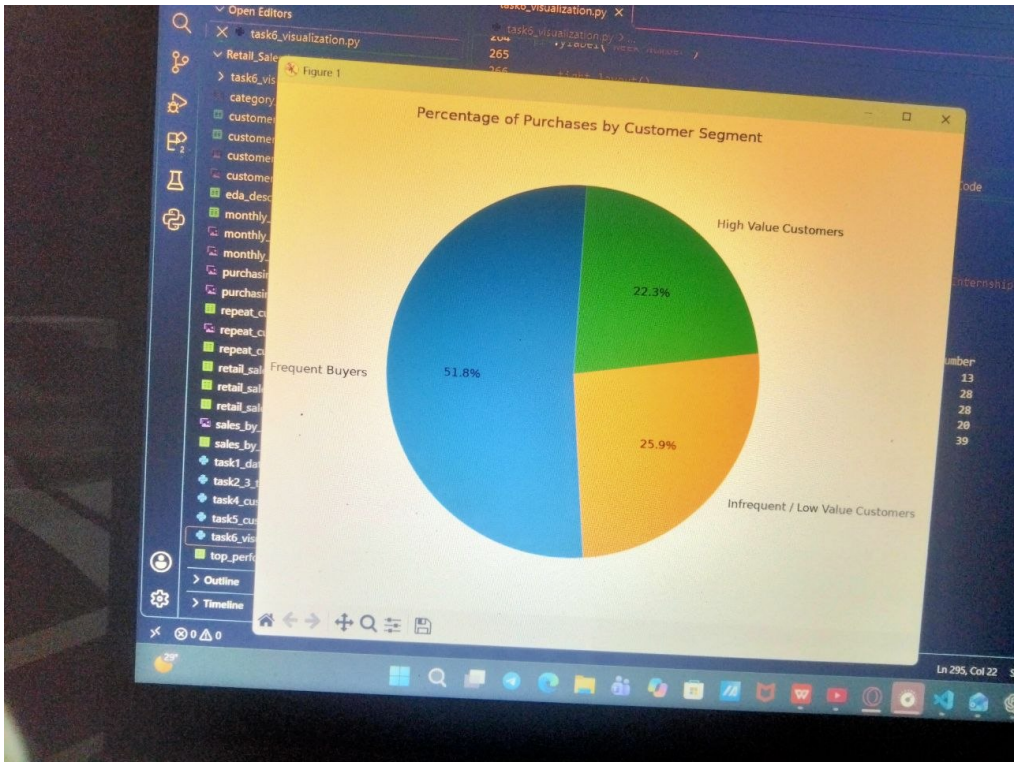



*Figure 12. Customer segment distribution.*

### Heatmap – Time-based category comparison

The heatmap compares category performance across time periods. More intense cells indicate relatively higher values and can help identify periods/categories for further investigation.

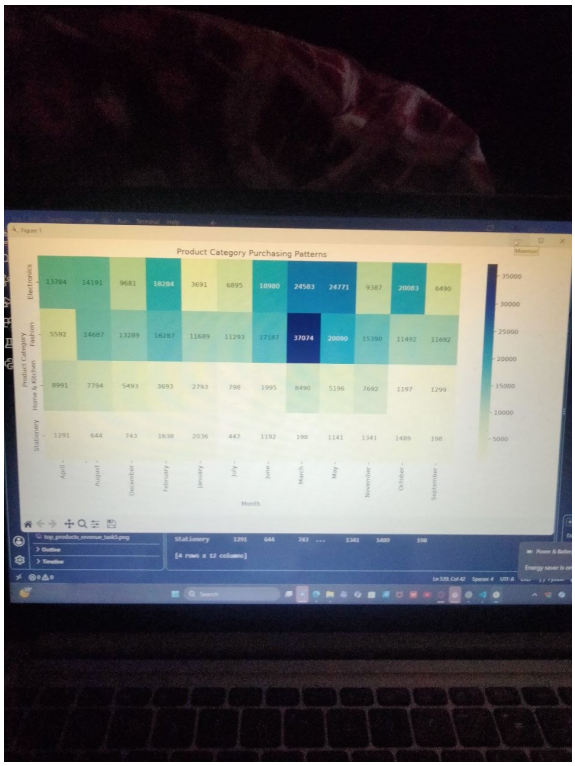



*Figure 13. Category-by-time heatmap.*

## 10. Overall Findings

| Area | Finding |
|---|---|
| Data quality | 253 original rows; 3 duplicates and 3 missing-Customer_ID rows removed; final 247 × 12; zero missing values. |
| Product quantity | Wireless Mouse led by quantity with 69 units. |
| Product revenue | Bluetooth Speaker led by revenue with ■99,942. |
| Monthly trend | March 2025 was strongest at approximately ■70,345. |
| Weekday trend | Friday was the strongest weekday in the reported comparison. |
| Customer segments | Frequent Buyers represented 51.8%. |
| Spending distribution | Most customers were in lower-to-middle spending ranges; a smaller group spent substantially more. |

# 11. Conclusion

The project demonstrates how Python can transform raw retail data into useful business information. The six tasks cover the analytical workflow from data quality checks to customer segmentation and visualization. The results can support inventory planning, product evaluation, customer retention and promotional timing. Further work may include profit-margin analysis, sales forecasting and market-basket analysis.

# 12. Project Deliverables

- Python scripts for the completed tasks.
- retail_sales_raw.csv – original dataset.
- retail_sales_cleaned.csv – cleaned dataset.
- retail_sales_transformed.csv – transformed dataset.
- EDA summary and generated CSV/PNG outputs.

Before submission, fill in the student and institution details on the cover page and ensure that the Python scripts, CSV datasets and chart outputs are included in the project folder or GitHub repository.